

6. Upgraded React Payload for Granular Defense

Overview

To feed the newly trained coverage-aware machine learning models, the frontend required a major upgrade to capture and transmit advanced defensive alignments. A simple "defense team" selection was no longer enough.

Code Changes

- **Frontend (React):** Added new `<select>` dropdowns in `situation.js` and `result.js` mapped to the `defenseCoverage` state variable.
- **Data Mapping:** Users can now select specific shells like `COVER_0`, `COVER_2`, `COVER_4`, and `2_MAN`. This state is bundled into the `current_situation` dictionary and sent to Flask, where it is mapped directly to the `defense_coverage_type` feature column the Scikit-Learn models expect.

Future Roadmap

Dropdowns are functional, but a visual interface is better. The roadmap includes building an interactive pre-snap canvas where users can literally drag and drop safeties and linebackers on the field. The React app will automatically calculate the box count and deduce the coverage shell based on the user's visual placements.